

LABORATORY

Lab 3

Radiation Detectors and Gamma Spectra · United States Air Force Academy, Department of
Physics and Meteorology

35 Points

NAMES

SCORE

/ 35

The purpose of this lab is to use detectors to determine various things about unknown radioactive sources. You will first use a Geiger counter to locate a source, and then you will use a scintillation detector to identify it.

I · BACKGROUND

A. Gamma-Ray Interactions with Matter

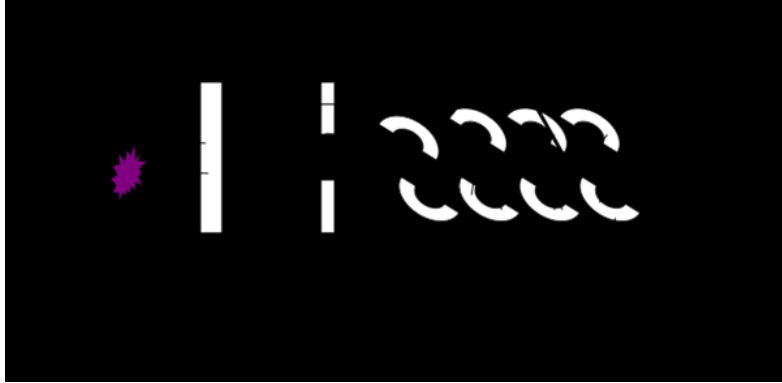
The three dominant methods gamma-rays interact with material are the photoelectric effect, Compton scattering, and electron-positron pair production.

In the photoelectric effect, the photon deposits 100% of its energy into an electron, freeing it from its bound state in an atom. When this occurs inside a detector, the output pulse produced will be proportional to the initial energy since all of the energy is transferred to the material.

In Compton scattering, the photon can scatter off of an electron transferring only a portion of its energy to the electron and retaining the rest. The amount of energy transferred depends on the scattering angle where the maximum energy occurs at 180° . If the photon does not go on to interact further with the detector, only a portion of the initial energy is measured by the detector. However, this maximum energy can produce a noticeable feature in scintillation detectors, known as the Compton edge, and can be calculated from the incident gamma ray energy. In pair production, an electron-positron pair are created when a gamma-ray interacts with a nucleus of an atom. These newly formed elementary particles slow down through scattering interactions, and, as the positron comes to rest, it annihilates with an electron producing a pair of back-to-back 511 keV gamma-rays. All of these interactions have the possibility of being measured by the detector. However, pair production can only occur if the photon has an energy greater than 1.022 MeV (the mass-energy of the electron-positron pair).

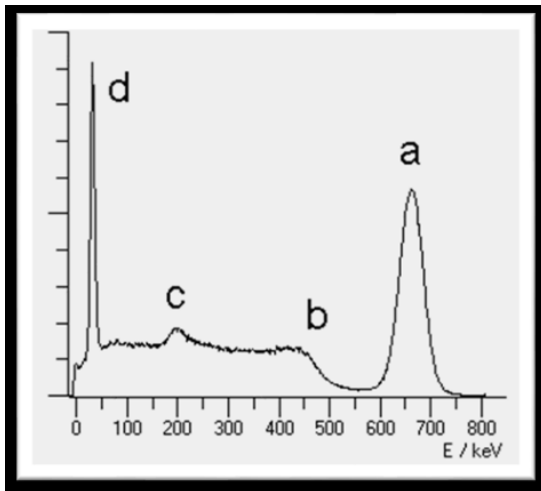
B. Scintillation Detectors

You have already worked with Geiger counters in previous labs. In this lab, we will also be working with a new type of detector — a scintillation detector. Scintillation detectors are created using materials that will scintillate, or produce light, when radiation deposits energy. They are one of the oldest types of radiation detectors, since measurements could be made with photographic film or the human eye. In modern scintillation detectors, light output is converted into current pulses using a photomultiplier tube as shown in the figure. The most common type of modern scintillators are made from thallium doped sodium iodide [NaI(Tl)] crystals which have ideal scintillation properties. That is what we'll be using today.



One of the most useful features of modern scintillation detectors is their ability to produce a light yield, and thus a current pulse, proportional to the energy deposited in the material by the radiation at a high efficiency. This proportionality provides the user with valuable information about the incident radiation's energy and therefore its source.

C. Gamma Spectra



FEATURE	CS-137 SIGNATURE	ENERGY
a	Gamma Decay Peak	662 keV
b	Compton Edge	??
c	Backscatter	184 keV
d	X-ray Peak	32 keV

In order to deduce meaningful information about a radiation source using a scintillation detector, many individual measurements are required. If each current pulse is measured, binned, and plotted as a histogram, features begin to emerge which can be very useful. For example, when we examine the figure above — if many current pulses are measured with the same magnitude, a large peak (a) is produced in the histogram. This tells us that one particular energy has been deposited in the scintillating material repeatedly, and likely indicates a radioactive source is present and producing radioactive particles with that very specific energy. Because every radioactive isotope has a unique decay mechanism, they each produce a unique energy signature.

These signatures act as an identifiable fingerprint for each isotope. By correlating the features revealed in the histogram from the scintillation detector with known energy signatures, unknown isotopes can be identified.

D. Energy Spectrum Calibration

The ability of scintillation detectors to produce these energy spectra are the primary characteristic that makes them one of the most useful detectors available. However, the spectra directly produced by the scintillation detectors are histograms of binned current pulse magnitudes (also known as channels), and they must be converted to energy spectra before they can be used. In order to convert the histogram from channels to energy, a calibration equation must be determined.

To accomplish this, a known source is measured with the detector first. We will use the Cs-137 source from the previous section as an example. Once a spectrum is gathered from the detector with the known source, a correlation can be drawn from the corresponding channels of each peak to the energies that are known to be produced by the known source. If channels for two features can be gathered from the histogram and correlated to the same features of the energy signatures from the isotope, the equation of a line can be found and used to convert any channel to its corresponding energy for future isotopes.

$$Energy = \frac{E_1 - E_2}{C_1 - C_2} * channel \quad (1)$$

E. Equipment

- Pancake counter (Geiger-Mueller detector)
- FLIR Identifinder (scintillation detector)
- FLIR Spectrum: <http://422278800004.local/App/Spectrum>

F. Procedure and Data

1. Cs-137 is a common calibration source for field detectors. A sample Cs-137 spectrum will be shown at the beginning of class.

KNOWN FEATURE	ENERGY	CHANNEL NUMBER
—	0 keV	0
X-ray Peak	32 keV	
Compton Edge		160
Primary Decay Peak	662 keV	220

Use the channel number of the primary peak as given and the channel number of the Compton edge along with Eq (1) to estimate the channel number of the 32 keV X-ray peak and the energy of the Compton edge.

6 pts

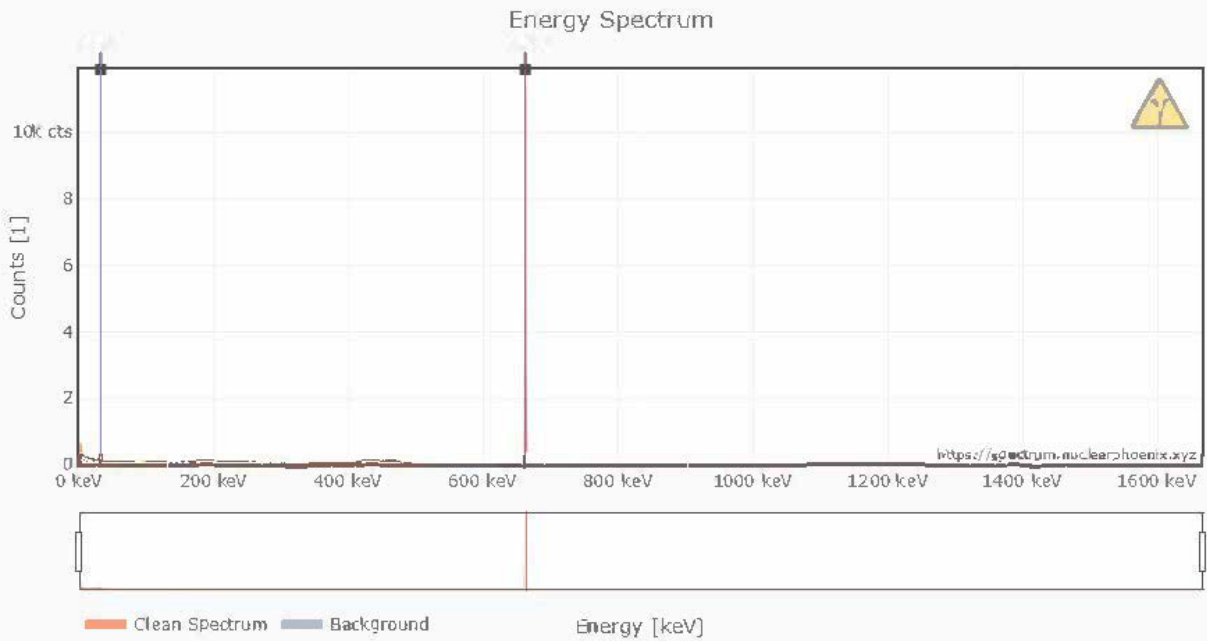
2. Get a pancake probe. Use it to survey the closed containers around the classroom and record the dose rate you measure at each. Do this until you find one that you believe contains a source (give a dose rate for at least 4 of the containers regardless of how quickly you find a source). Once you have picked one that you believe contains a source, alert your instructor who will open the box.

3 pts

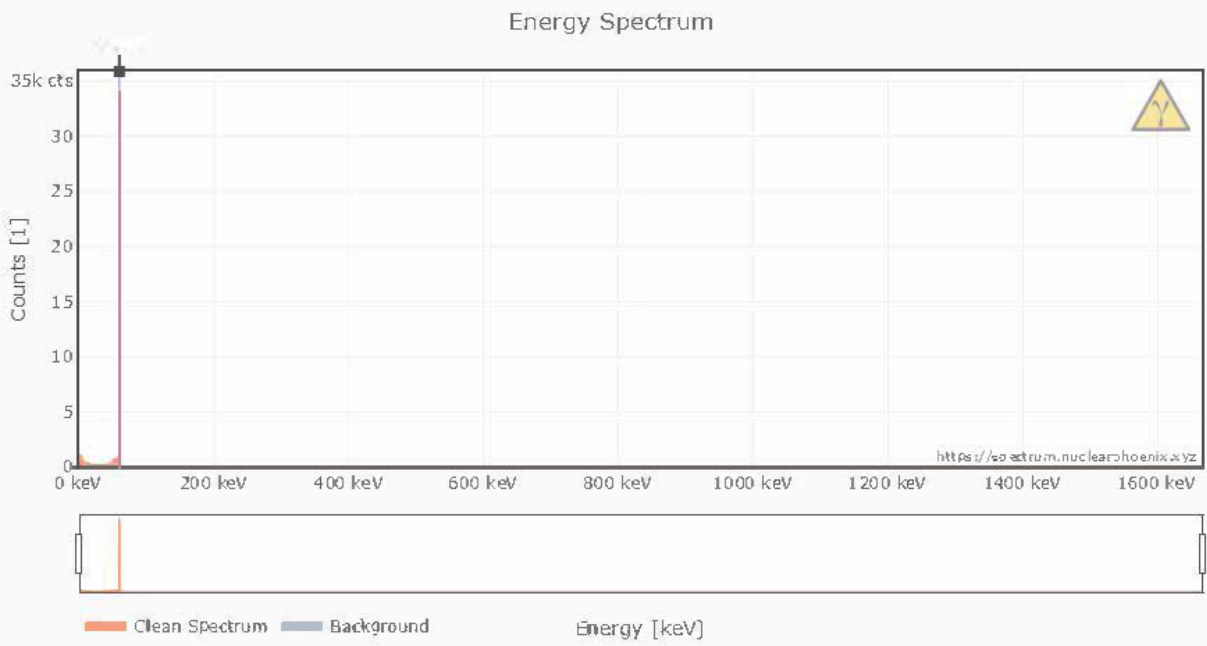
CONTAINER DESCRIPTION	MEASURED DOSE RATE

3. Analyze the spectra shown below for 5 different isotopes. Use your equation sheet to determine what spectra corresponds to what isotope. [10 pts]

Isotopes to consider: Na-24, K-40, Co-60, Mo-99, I-131, Xe-135, Cs-137, Am-241.

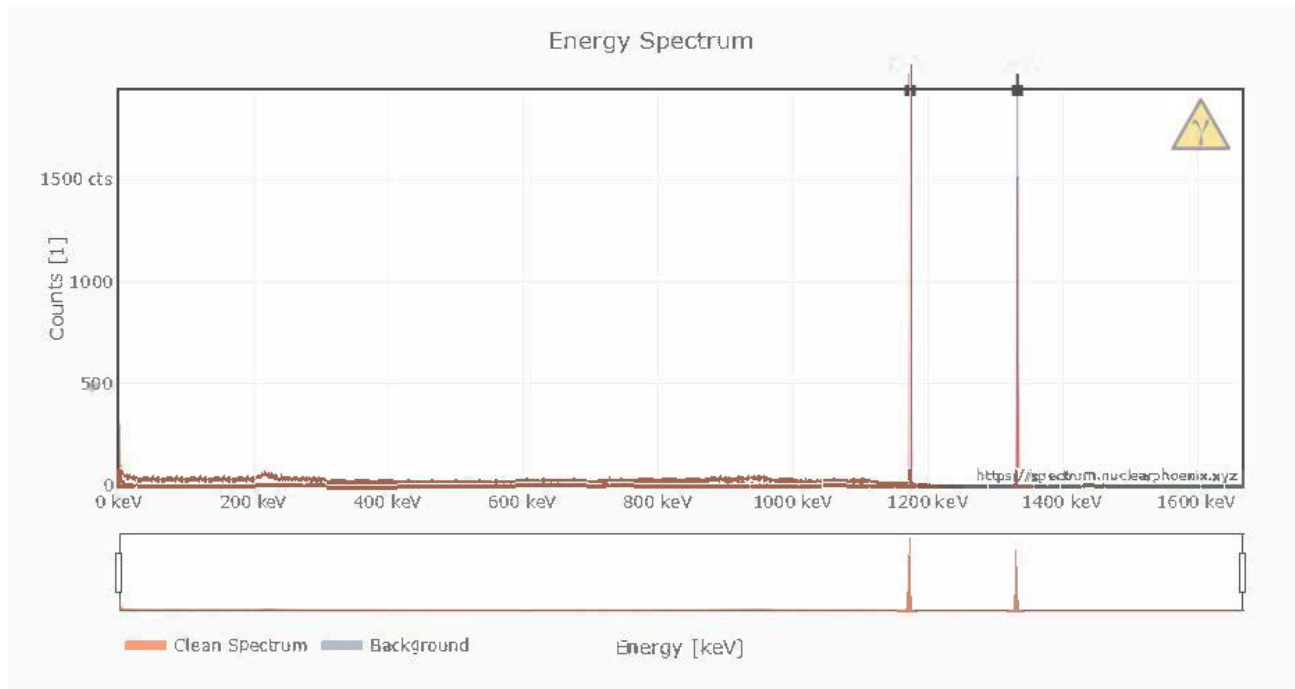


Isotope: _____



Isotope: _____





Isotope: _____

G · DISCUSSION QUESTIONS

5 pts **Question 1.** For the final spectrum, the two primary decay peaks were found at channels 390 and 442. Use Eq (1) to numerically determine the energies of these peaks.

5 pts

Question 2. The equation normally used to calculate the energy of the Compton edge is

$$E_{\text{compton}} = \frac{2E^2}{m_e c^2 + 2E}$$

where E is the energy of the initial gamma, and $m_e c^2$ is the rest energy of the electron. Calculate the Compton edge energy using the energy calculated above for the primary peak.

3 pts

Question 3. Discuss the importance of scintillation detectors in radiation safety and nuclear science. What benefits do these detectors have over the Geiger-Mueller tubes?

3 pts

Question 4. That said, your pancake counter (a gas-filled detector) was also useful in this lab. What are some advantages of gas counters? Identify a situation in which you might prefer one of these detectors over a scintillator or solid-state detector.